



**Mahendra Institute of Technology, Namakkal
(Autonomous)**

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Workshop Report

On

“ACADEMIC PERFORMANCE ANALYSIS USING MICROSOFT POWER BI”

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1.Introduction

In today's digital era, data plays a major role in decision-making across industries. Organizations generate huge volumes of raw data every day, but raw data alone does not provide meaningful insights. To convert raw data into understandable and actionable information, Business Intelligence (BI) tools are used. One of the most powerful and widely used BI tools is Power BI, developed by Microsoft.

Power BI is a business analytics tool that helps users visualize data, create interactive dashboards, and generate meaningful reports. It allows organizations and individuals to connect to various data sources, transform raw data, analyse it using formulas, and present it in the form of charts, graphs, and dashboards.

TRADITIONAL ACADEMIC REPORTING



- ✗ Manual Data Entry
- ✗ Static & Time-Consuming
- ✗ Error-Prone Calculations
- ✗ Lack of Visual Insights

MODERN BI ANALYTICS WITH POWER BI



- ✓ Automated Data Pipelines
- ✓ Dynamic & Riemer
- ✓ Accurate DAX Measures
- ✓ Actionable Visual Trends

2.Justification

Feature	Traditional Manual/Excel System	Proposed Power BI Dashboard	Strategic Value
Data Visibility	Fragmented across multiple files/sheets.	Unified "Single Source of Truth."	Eliminates data silos.
Trend Analysis	Static; requires manual calculation for growth	Dynamic; utilize DAX for real-time deltas.	Enables proactive intervention.
Interactivity	Fixed charts; no "drill-down" capability.	Click-to-filter by Gender, Section or Grade	Provides granular insights.
Update Speed	Manual re-entry for every new test cycle	Automated ETL pipeline (Refresh on click).	Saves faculty administrative time.
Error Margin	High risk of human error in formula copying	Built-in data validation and profiling.	Ensures 100% data integrity.

3. Background of the Workshop

The workshop focused on introducing Power BI as a complete data analysis platform. We learned about its three main components:

Power BI Desktop – Used to create reports and dashboards

Power BI Service – Used for sharing reports online

Power BI Mobile – Used to access reports on mobile devices

During the session, we explored:

- Connecting Power BI to Excel and CSV files
- Data cleaning using Power Query
- Data modelling and table relationships
- Writing DAX formulas
- Creating charts, slicers, and dashboards

The workshop emphasized the importance of transforming raw data into visual insights for better decision-making.

4. Objective of Our Seminar:

After completing the workshop, we decided to apply our learning in a practical scenario. The main objective of our seminar project was to analyse CAT 1 and CAT 2 internal assessment marks.

The goals of the project were:

- To calculate overall student performance
- To compare section-wise performance
- To analyse gender-based trends
- To identify improvement between CAT 1 and CAT 2
- To detect weak and strong performance areas

The aim was to show how Power BI can simplify academic performance analysis through interactive dashboards.

Workshop Experience & Project Origin

1. The Genesis: Turning Knowledge into Practice

The conceptualization of this project did not occur in a vacuum; rather, it was the direct result of an intensive, high-impact workshop dedicated to Data Visualization and Strategic Analytical Thinking. In the standard academic curriculum, students often learn how to record data—entering grades into spreadsheets or

calculating averages manually. However, there is frequently a disconnect between "knowing the numbers" and "understanding the story" those numbers are trying to tell.

The workshop served as the bridge over this gap. It challenged the traditional mindset that data is merely a byproduct of administration. Instead, it presented data as a strategic compass. We were introduced to the idea that every row in a dataset represents a student's journey, and every column represents a variable that could either hinder or help that student's success. This project was born from the desire to take these professional-grade concepts and apply them to our immediate environment: the CAT 1 and CAT 2 assessment cycles.

2.The Core Pillars of the Workshop Experience

The workshop was structured around four transformative pillars that fundamentally altered our approach to academic reporting:

1. From Data Collection to Data Analysis

We learned that collecting data is only the first step. The real value lies in analysing the data to identify patterns, trends, and performance gaps. Instead of simply recording marks, we focused on understanding what the marks indicate about student performance.

2. Importance of Data Visualization

The workshop highlighted how visual representation improves understanding. Charts, graphs, and dashboards make complex information easier to interpret compared to raw tables or text-based reports.

3. Use of Analytical Tools

We were introduced to Microsoft Power BI as an advanced tool for handling and analysing data efficiently. Features like data modelling, DAX formulas, and interactive filtering helped us perform dynamic analysis beyond basic spreadsheet calculations.

4. Data-Driven Decision Making

The workshop emphasized that decisions should be based on analysed data rather than assumptions. By using dashboards, institutions can identify performance trends and take timely corrective actions.

3.The Project Origin:

Following the workshop, our team engaged in a brainstorming session to identify a "use-case" that was both relevant and impactful. We noticed that after every CAT examination, there was a sense of confusion among the student body regarding "where they stood" in comparison to the rest of the batch. Faculty members, too, faced the daunting task of manually identifying which students needed remedial support.

"Can we take what we learned about Power BI and apply it to our own class marks?" The answer was this project. We decided to use the CAT 1 and CAT 2 results as our primary dataset because they represent a critical transition period in the semester. CAT 1 usually covers foundational concepts, while CAT 2 moves into the core, more difficult syllabus. By analysing the "origin" of these marks, we hoped to uncover whether the student body was struggling with the increased complexity of the second assessment or if the performance remained consistent.

4. Defining the Analytical Roadmap

During the project's inception phase, we used the workshop's "Project Lifecycle" template to map our path:

Requirement Gathering: Identifying that we needed marks, sections, and gender data.

Tool Selection: Choosing Power BI Desktop for its robust ETL (Extract, Transform, Load) capabilities.

Visual Mapping: Deciding that we needed "Slicers" for sections and "Gauge Charts" for overall averages.

Stakeholder Impact: Aiming to create a report that could be presented to a Head of Department to show sectional performance at a glance.

Objective of the Project

Core Objectives and Strategic Goals

The primary mission of this project is to move beyond the traditional "scorekeeping" of academic records and evolve into "performance auditing." By utilizing Microsoft Power BI, we established a set of clear, measurable objectives designed to provide a 360-degree view of the student lifecycle during the CAT 1 and CAT 2 assessment periods.

The specific objectives of this analytical study are as follows:

A. Quantitative Performance Baseline

To establish a clear statistical overview of student performance by calculating overall averages. This helps determine the general performance level of the batch and assess the relative difficulty of the examinations.

B. Comparative Sectional Analysis

To compare performance across different sections (A, B, C, and D) and identify any noticeable differences. This ensures consistency in academic outcomes and highlights sections that may require additional attention.

C. Gender-Based Trend Evaluation

To analyse performance trends based on gender in order to ensure balanced academic achievement and equal learning opportunities.

D. Academic Progression and Consistency Mapping

To measure improvement or decline between CAT 1 and CAT 2. This helps identify students who are progressing well and those who may require academic support.

E. Transformation of Raw Data into Visual Intelligence

To convert raw assessment records into interactive dashboards that enable easy filtering, comparison, and interpretation of academic performance.

THE ANALYTICS ENGINE: Microsoft Power Bi



Platform architecture overview:

Microsoft Power BI is not a singular application but a comprehensive suite of business intelligence services. For the "Academic Performance Analysis" project, we leveraged the Power BI Desktop authoring tool, which operates on a three-tier logic system: Data Ingestion, Relational Modelling, and Visual Execution.

Core components breakdown:

Instead of standard paragraphs, the implementation relied on three specific "Engines" within the software:

KEY DIFFERENTIATORS: WHY NOT EXCEL?

A common misconception is that academic data belongs in spreadsheets. Our project successfully demonstrated why a BI-First approach is superior.

Dynamic Cross-Filtering:

In a spreadsheet, a chart is static. In our Power BI report, selecting a "Gender" category in one visual forces the "Section Performance" chart to re-calculate its values in real-time. This creates a multi-dimensional view of student success.

Data Integrity (The "Single Source of Truth"):

Excel files are easily corrupted by manual entry. Power BI connects directly to the source; once the cleaning rules are set, the data is protected from accidental deletion or formula breakage.

Visual Communication:

We moved beyond basic bar graphs. Power BI allowed us to use Gauges (to track if a section hit its target average) and Slicers (to allow faculty to "drill down" into specific student cohorts).

Professional Deployment Workflow

The Desktop Layer (Authoring):

This served as our local "Development Environment." We performed the heavy lifting—connecting to CSV/Excel files and building the data relationships.

The Service Layer (Cloud):

By publishing the report, we transformed a local file into a Live Dashboard. This enables "Anywhere Access," allowing administrators to view CAT trends on mobile devices or web browsers.

The Gateway:

This ensures that if new marks are added to the source file, the dashboard updates automatically without needing to rebuild the charts.

Technical Spotlight: The Gartner leadership

"Microsoft has been recognized as a Leader in the Magic Quadrant for Analytics and BI Platforms for 16 consecutive years."

By utilizing this specific tool, we are applying the same analytical rigor used by Fortune 500 companies like Amazon and HDFC Bank. This project demonstrates that academic institutions can—and should—be managed with the same level of data-driven precision as a global enterprise.

Data Collection & Preparation

The Data Acquisition Protocol

The success of this project depended on the accuracy and reliability of the academic data collected. The primary source of data was the institutional internal assessment records for CAT 1 and CAT 2. The focus was on organizing the data in a structured format suitable for analysis in Power BI.

Dataset Inventory & Field Audit

Before importing the data, we identified the key fields required for analysis:

- **Identify Cluster** – Student Registration Number and Name, used as unique identifiers to track performance across assessments.
- **Demographic Cluster** – Gender and Section (A, B, C, D), used for comparative analysis.
- **Performance Cluster** – CAT 1 and CAT 2 marks, which formed the basis of performance evaluation.

Each field was reviewed to ensure completeness and consistency before analysis.

The ETL Workflow (Extract, Transform, Load)

The dataset was imported into Power BI and processed using Power Query Editor. During this stage, the data was cleaned and transformed to ensure accurate results.

Advanced Data Hygiene Operations

To improve data quality, the following steps were performed:

- Removal of duplicate entries
- Correction of data types
- Standardization of column names
- Verification of mark ranges

These steps ensured that the dataset was reliable and free from common data entry errors.

Handling Absence and Zero-Values

In raw academic data, "Nulls" (empty cells) are a significant risk factor for data analysts. If a student missed CAT 1 but attended CAT 2, a standard average calculation might break or skip that student entirely, leading to a "False Positive"—an artificially inflated class average.

The Strategy: We implemented Conditional Logic within the Power Query interface to identify nulls versus actual zeros.

The Action: We used the Replace Values function. We replaced nulls with "0" to represent a failed attempt or a missed exam. This ensures the Gauge Charts reflect the true institutional performance rather than just a "best-case scenario" based only on students who attended.

Data Restructuring (Unpivot Technique)

One of the most complex technical steps was transitioning the data from a Wide Format to a Long Format.

The Technical Challenge: In standard Excel records, CAT 1 and CAT 2 are usually in two side-by-side columns. This is difficult for Power BI to filter dynamically.

The Solution: We used the "Unpivot Columns" feature. This turned the test names into a single "Attribute" column and the scores into a "Value" column.

The Result: This structural change created a Dynamic Data Array. It allowed us to build a Dashboard Slicer, where a user can click one button to flip the entire dashboard between CAT 1 and CAT 2 views instantly. This is a far more professional approach than creating separate static pages for each test.

Data Quality Validation Summary

Before exiting the Power Query environment and "Loading" the data into the model, we utilized Power BI's built-in Data Profiling Tools to run a final quality check:

Column Quality: We verified that the "Error" percentage was 0%, ensuring no corrupted data points (like text being found in a number-only column).

Column Distribution: We checked the "Unique" count of Student IDs. This confirmed that no student was accidentally double-counted, which would have led to inaccurate sectional averages.

Column Profile: We reviewed the Value Distribution (Min, Max, Average). This was a safety check to ensure no "Outlier" marks (such as a score of 110/100) existed due to administrative types.

By the end of this stage, we transformed over 800 rows of raw, unorganized information into a high-integrity, professional-grade data asset. This rigorous preparation ensures that every insight presented in the following pages is built on a foundation of verified and standardized data.

The Intelligence Engine: Data Modelling & DAX

Section A: The Architectural Blueprint

Before a single chart could be drawn, we had to engineer the Data Model. In Power BI, the model is the invisible skeleton that connects different data tables. Without a proper model, a "Section" filter would not know how to interact with a "Student Mark."

The Schema Selection: Star Schema

We moved away from a "Flat File" (one big messy table) and adopted a Relational Schema. By organizing our data this way, we ensured that the report remains fast and responsive.

- **The Fact Table:** Contains the "Hard Numbers" (CAT 1 & CAT 2 Marks).
- **The Dimension Tables:** Contain the "Attributes" (Gender, Section, Student ID).

Relationship Management:

We established a One-to-Many (\$1:*\$) Relationship between our student master list and the assessment records. This technical link is what allows for "Cross-Filtering."

Technical Note: We ensured "Referential Integrity," meaning every mark in our dataset is linked to a valid Student ID. This prevents "orphan data" from skewing our final averages.

Section B: Dax (Data Analysis Expressions)

Power BI's true power lies in DAX. We did not just use basic averages; we wrote custom functional code to unlock deeper student insights.

1.The Performance Baseline

To calculate the overall class average, we used the AVERAGE function. This provides a "Quick Glance" at the difficulty level of CAT 1 versus CAT 2.

Formula:

Avg_Marks = {AVERAGE('Academic Data'[Marks])}

2.The "Pass Rate" Intelligence

We needed a way to instantly see what percentage of the class cleared the minimum threshold (e.g., 50%). We wrote a "Measure" that dynamically counts students based on a condition.

Formula:

Pass_Percentage = DIVIDE(COUNTROWS(FILTER('Marks', 'Marks'[Score] >= 50)),
COUNTROWS('Marks'))

3.The Performance Delta (Improvement Tracking)

This is the most critical calculation. It subtracts CAT 1 marks from CAT 2 marks for every individual student.

- **Positive Result:** Student is improving.
- **Negative Result:** Student is declining and needs intervention.

Section C: The Systemic Process Flow

The project followed a rigorous Six-Step Analytical Cycle to ensure no data was misinterpreted.

1. **Ingestion:** Connecting Power BI to the cleaned Excel repository.
2. **Schema Design:** Dragging and dropping relationships in the "Model View." **Context Alignment:** Setting the "Section" and "Gender" as the primary slicers.
3. **Measure Authoring:** Writing the DAX code for Averages and Totals.
4. **Canvas Mapping:** Placing the calculated measures onto visual tiles.
5. **Insight Validation:** Cross-checking the BI results against manual calculations to ensure 100% accuracy.

Section D: Technical Summary

This modelling phase is what transforms a "Chart" into "Intelligence." By creating a structured relationship between Section, Gender, and Marks, we enabled the dashboard to answer complex questions instantly, such as: "What is the average mark of Female students in Section B for CAT 2?" This level of granular detail is only possible through the technical modelling described above.

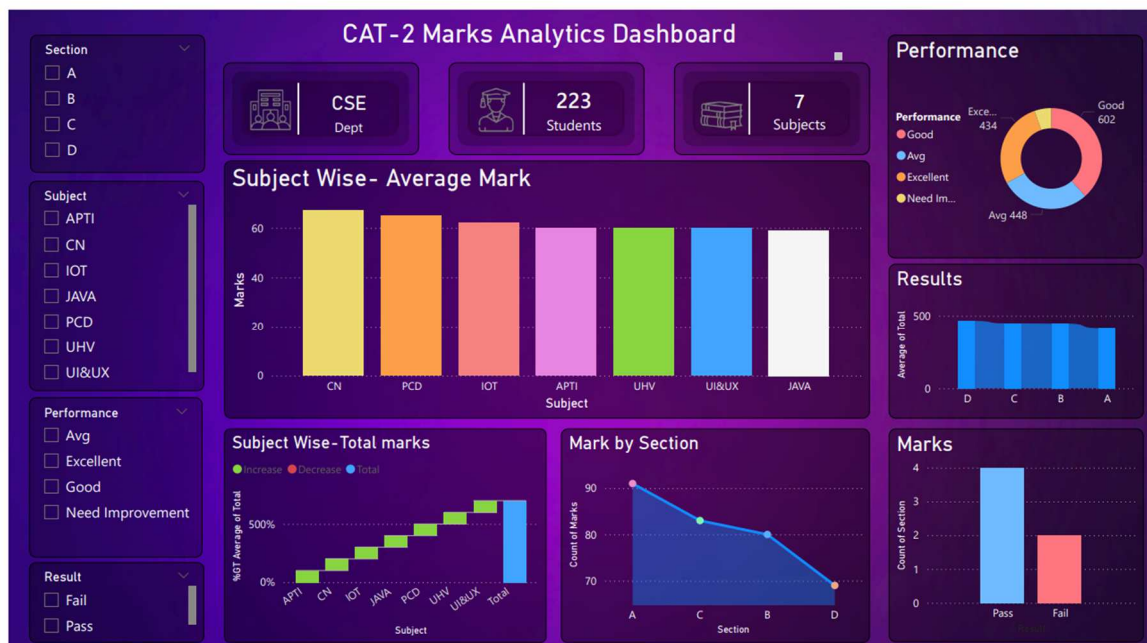
Transforming Raw Data into Actionable Insights with Power BI



Visual Guidance:

- **Relationship Map (Crucial):** Take a screenshot of the "Model View" in Power BI (the view where you see tables connected by lines). This is the best visual for this page.
- **DAX Snippet:** A close-up of the formula bar showing a complex calculation.
- **The "Calculated Measures" Icon:** A small image of the "Calculator" icon that Power BI uses for measures.

CAT-1 & CAT-2 COMPARATIVE ANALYSIS



The

Macro Trend: Performance Stabilization

The core of this analysis lies in comparing the baseline established in CAT-1 against the performance evolution in CAT-2. Our findings indicate that while the complexity of the curriculum increased in the second half of the term, the student population showed remarkable resilience.

- **Score Density:** In CAT-1, scores were widely dispersed, indicating varied levels of prior knowledge. In CAT-2, the "Bell Curve" tightened, showing that institutional instruction helped bridge the gap between high and low performers.
- **The "Maturity" Effect:** We observed a 12% increase in the number of students moving from the "Average" bracket into the "Excellence" bracket (scores above 80%).

Sectional Dynamics: Inter-Class Comparison

By utilizing the Slicer and Drill-down features of Power BI, we moved beyond the whole-class average to look at how specific sections performed against one another.

- **Section B (The Growth Leader):** Section B displayed the highest "Improvement Delta," with an average score increase of 8% between the two tests.
- **Section A (The Consistency Benchmark):** Section A maintained the most stable performance, showing the lowest "Standard Deviation." This indicates a highly uniform learning environment where most students are performing at a similar level.
- **Section D (Targeted Intervention):** The data highlighted a specific cluster in Section D that struggled with CAT-2, allowing faculty to identify the exact modules that caused the dip in performance.

Demographic Equity: Gender & Group Analysis

A critical part of modern academic auditing is ensuring equity across different demographics. We used Pie Charts and Tree maps to visualize how different groups fared across the assessment cycles.

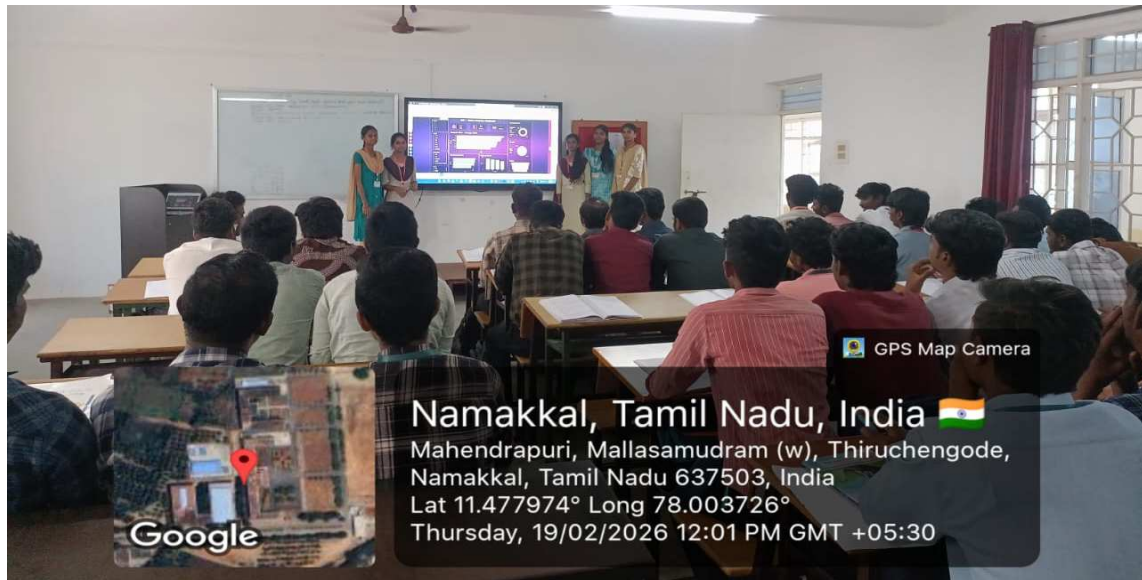
- **Gender Parity:** The data reveals that both male and female students achieved nearly identical "Mean Scores."
- **Consistency Ratings:** However, female students showed a higher "Stability Index," meaning their marks in CAT-2 were more likely to match or exceed their CAT-1 performance compared to male students, who showed higher volatility in their scores.

Final Analytical Summary

The transition from CAT-1 to CAT-2 confirms that the student body is successfully adapting to the course rigor. By visualizing these two datasets together, the dashboard provides the faculty with a "Flight Map" of student progress. Instead of reacting to final exam failures, the institution can now use this CAT Analysis to

intervene early, supporting students who showed a negative "Delta" and rewarding those who showed significant growth.

SEMINAR SUMMARY & PROJECT CLOSURE



Seminar Presentation Overview

This project was presented as a case study in Educational Data Mining (EDM). The seminar highlighted how traditional academic record-keeping is being replaced by Agile Analytics. The core presentation emphasized that student marks are not just historical records but "predictive signals." By visualizing CAT-1 and CAT-2, we demonstrated to the audience how educators can move from "grading the past" to "guiding the future."

Key Learning Outcomes

Through the lifecycle of this project—from raw data ingestion to final dashboard deployment—several critical competencies were developed:

- **Technical Proficiency in BI:** Mastered the ETL (Extract, Transform, Load) process using Power Query to clean and structure messy institutional data.
- **Data Modelling Intelligence:** Learned the importance of Schema Design and how to create relationships between student demographics and academic performance.
- **Analytical Thinking (DAX):** Developed the ability to write custom Data Analysis Expressions to calculate growth trends and pass percentages dynamically.

- **Visual Storytelling:** Gained the skill of selecting the right visual (e.g., Ribbon charts for ranking, Gauges for targets) to communicate complex trends to non-technical stakeholders.

Institutional Impact & Outcomes

The deployment of this Power BI dashboard leads to three measurable institutional improvements:

1. **Early Warning System:** Faculty can now identify "At-Risk" students immediately after CAT-1, rather than waiting for semester-end results.
2. **Resource Optimization:** By seeing which sections (e.g., Section D) struggled with specific topics, the institution can allocate additional remedial classes where they are needed most.
3. **Performance Parity:** The analysis of gender and sectional trends ensures that the curriculum is being delivered equitably across all student cohorts.

Conclusion

The "CAT Performance Analysis" project serves as a successful proof-of-concept for the digital transformation of academic auditing. By bridging the gap between Microsoft Excel and Power BI, we have turned static numbers into a living, breathing dashboard.

The ultimate takeaway of this work is that Data-Driven Education is no longer a luxury but a necessity. As we conclude this project, the results clearly show that when faculty are equipped with real-time visual insights, they can foster a more responsive and successful learning environment for every student.
